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**B.Sc. (E.C.S.) (Semester - II) (NEP CBCS) Examination:
October/November – 2025**

Introduction to Microcontroller and Embedded System (G06-0203)

Day & Date: Thursday, 13-11-2025
Time: 03:00 PM To 04:30 PM

Max. Marks: 30

Instructions: 1) All questions are compulsory.
2) Each question should only one correct option.
3) Figure to right indicate full marks.

Q.1 Multiple Choice Question:

06

- 1) 8051 is _____ bit microcontroller.

a) 8	b) 16
c) 4	d) 19
- 2) Maximum external memory can be connect to 8051_____.

a) 64k	b) 8k
c) 128 byte	d) 4k
- 3) To access external memory pin EA connect to _____.

a) gnd	b) +vcc
c) None	d) Both a and b
- 4) _____port required Pull-Up register.

a) Port 0	b) Port 1
c) Port 2	d) Port 3
- 5) _____ port is a mono function port.

a) 0	b) 1
c) 2	d) 3
- 6) _____ port used as lower address port.

a) 0	b) 1
c) 2	d) 3

Q.2 Answer the following questions. (Any Three)

06

- a) Draw Basic architecture of embedded system.
- b) Define Addressing Mode.
- c) State application of 8051 embedded system.
- d) Write Data Transfer instructions of microcontroller.
- e) Write ALP to add two 8 bit numbers

- Q.3 Answer the following questions. (Any Two) 06**
- a) Alternative function of Port 3.
 - b) Internal memory of 8051.
 - c) Flash magic tools.
- Q.4 Answer the following questions. (Any Two) 06**
- a) Explain types of embedded system.
 - b) Differentiate between microprocessor and microcontroller.
 - c) Write features of 8051.
- Q.5 Answer the following question. (Any One) 06**
- a) Draw and explain a block diagram of 8051.
 - b) Write different Instruction types of microcontrollers and explain in detail.